

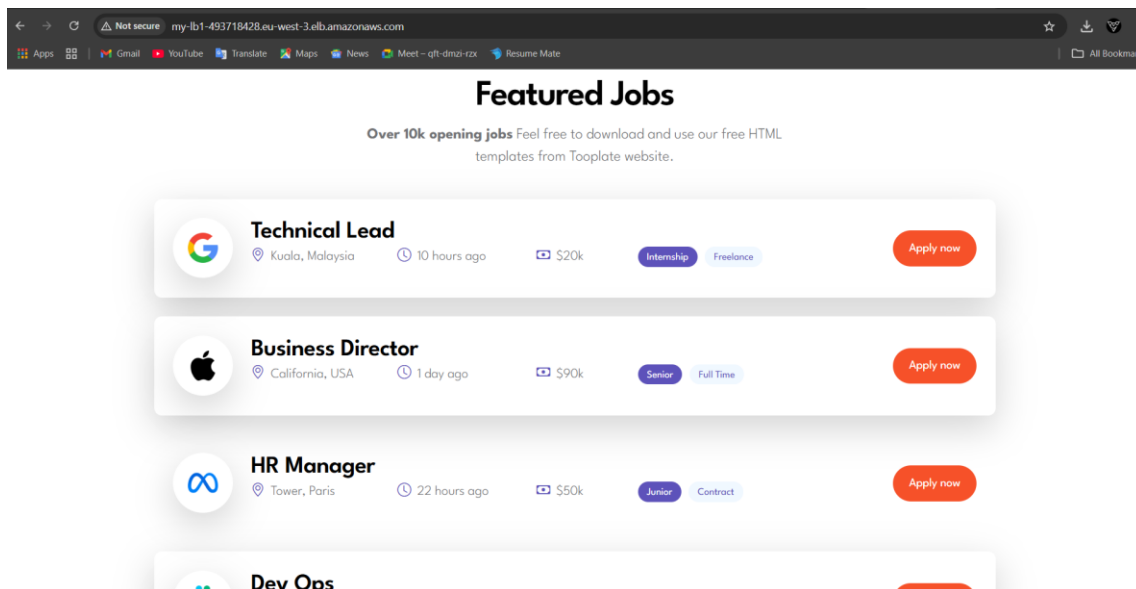
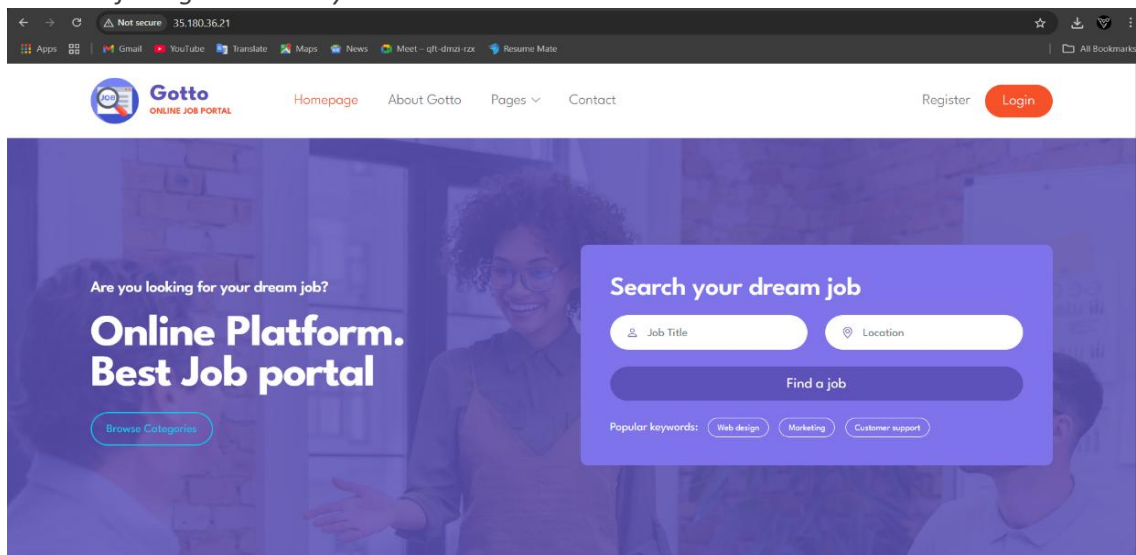
AWS PROJECT

Project Overview

Website hosting on AWS by creating a custom VPC with Auto Scaling, Application Load Balancer, and monitoring using CloudWatch. Also integrates Simple Notification Service (SNS) for alerts and notifications. And also created Private Database.

Web server with Public IP & Load Balancer DNS

Deployed a Clean Work HTML template on EC2 with a public IP and integrated it behind an Application Load Balancer for high availability.



EC2 INSTANCES

Launched EC2 instances for the project using Auto Scaling to ensure high availability and scalability.

The screenshot displays the AWS Management Console for EC2 instances. The top navigation bar shows the AWS logo, a search bar, and the user's name 'omkar'. The main header indicates 'EC2 > Instances'. Below this, there's a section for 'Instances (4/4)' with a filter set to 'Running'. A table lists four instances: 'database-server', 'webserver1', 'webserver2', and 'jump-server1'. Each instance is in a 'Running' state, using a 't2.micro' instance type, and is located in the 'eu-west-3a' availability zone. The table also shows status checks, alarm status, and public IP addresses. At the bottom, there's a 'Monitoring' section with a 'Configure CloudWatch agent' button.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
database-server	i-00167c533a6e24682	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-3a	-	-
webserver1	i-0ea06fa41a93a6cab	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-3a	-	35.180.36.21
webserver2	i-03e229b4c254ff4eb	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-3a	-	15.237.254.153
jump-server1	i-0f6f0356fa854802b	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-3a	-	13.38.80.178

VPC

Created a custom VPC to securely manage and isolate resources for my project within AWS.

The screenshot shows the AWS Management Console for VPCs. The top navigation bar is the same as the EC2 page. The main header indicates 'VPC > Your VPCs'. Below this, there's a section for 'Your VPCs (1/2)' with a filter set to 'Available'. A table lists two VPCs: 'my-project-vpc' and an unnamed VPC. The 'my-project-vpc' is in an 'Available' state, has a 'Block Public Access' setting of 'Off', and is associated with the '10.10.0.0/16' IP address. The unnamed VPC is also in an 'Available' state, has a 'Block Public Access' setting of 'Off', and is associated with the '172.31.0.0/16' IP address. Below the table, there's a section for 'vpc-01b5ccf52853e0c53 / my-project-vpc' with tabs for 'Details', 'Resource map', 'CIDRs', 'Flow logs', 'Tags', and 'Integrations'. The 'Details' tab is selected, showing the VPC's details.

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set
my-project-vpc	vpc-01b5ccf52853e0c53	Available	Off	10.10.0.0/16	-	dopt-028591ce871bab...
-	vpc-079dda54e70ccb3bb	Available	Off	172.31.0.0/16	-	dopt-028591ce871bab...

Subnets

Configured public and private subnets within the VPC to separate internet-facing and internal resources.

aws

Search

[Alt+S]

Europe (Paris)

omkar

VPC

Subnets

Subnets (3/6)

Info

Last updated about 2 hours ago

Actions

Create subnet

Find subnets by attribute or tag

☒	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☒	public-subnet-project	subnet-0ac05596834545e7e	Available	vpc-01b5ccf52853e0c53 my-p...	Off	10.10.1.0/24	-
☒	private-subnet-project	subnet-0c1c34325285f7763	Available	vpc-01b5ccf52853e0c53 my-p...	Off	10.10.2.0/24	-
☐	-	subnet-01a64c7b94cd91b98	Available	vpc-079dda54e70ccb3bb	Off	172.31.0.0/20	-
☐	-	subnet-0a747a59cb5a230ea	Available	vpc-079dda54e70ccb3bb	Off	172.31.32.0/20	-
☐	-	subnet-03308187653d9621e	Available	vpc-079dda54e70ccb3bb	Off	172.31.16.0/20	-
☒	public-subnet-dummy	subnet-034bb307ddd45b772	Available	vpc-01b5ccf52853e0c53 my-p...	Off	10.10.3.0/24	-

Subnets: subnet-0ac05596834545e7e, subnet-0c1c34325285f7763, subnet-034bb307ddd45b772

Route Tables

Created and associated route tables to control traffic flow between subnets and external networks.

aws

Search

[Alt+S]

Europe (Paris)

omkar

VPC

Route tables

Route tables (2/5)

Info

Last updated 1 minute ago

Actions

Create route table

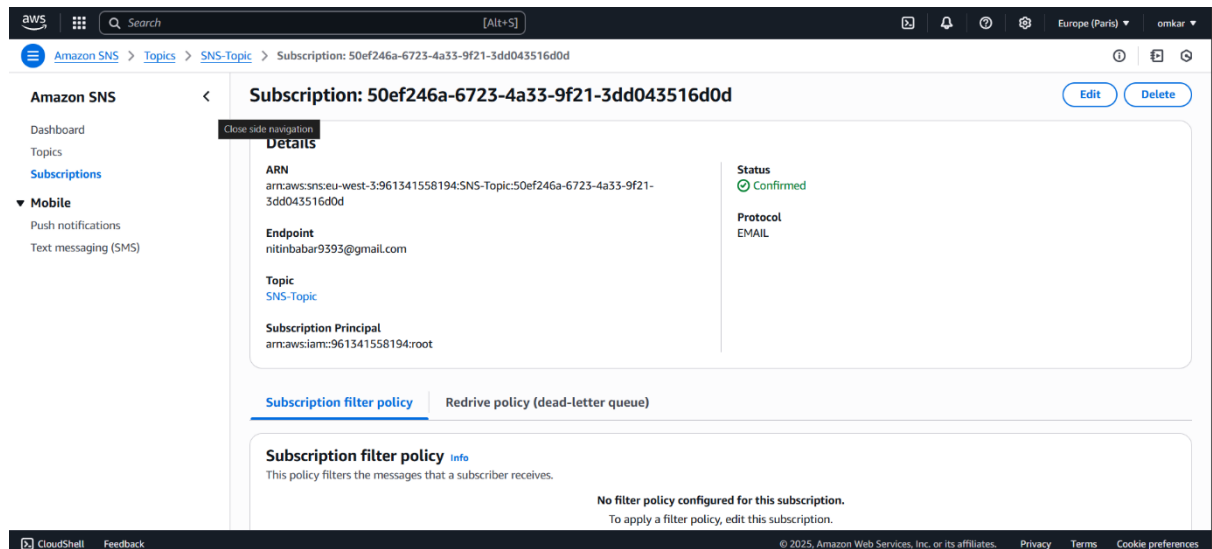
Find route tables by attribute or tag

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☒	my-private-RT	rtb-0b80e93c75da061eb	subnet-0c1c34325285f7...	-	No	vpc-01b5ccf52853e0c53 my-p...	961341558194
☐	-	rtb-0716be804e399f9f2	-	-	Yes	vpc-01b5ccf52853e0c53 my-p...	961341558194
☒	my-public-RT	rtb-0561aabd22b355beeb	subnet-0ac05596834545...	-	No	vpc-01b5ccf52853e0c53 my-p...	961341558194
☐	-	rtb-0e6a5f9172356a593	-	-	Yes	vpc-079dda54e70ccb3bb	961341558194
☐	RDS-Pvt-rt	rtb-0134c14e2fce0fd1b	-	-	No	vpc-01b5ccf52853e0c53 my-p...	961341558194

Route tables: rtb-0b80e93c75da061eb, rtb-0561aabd22b355beeb

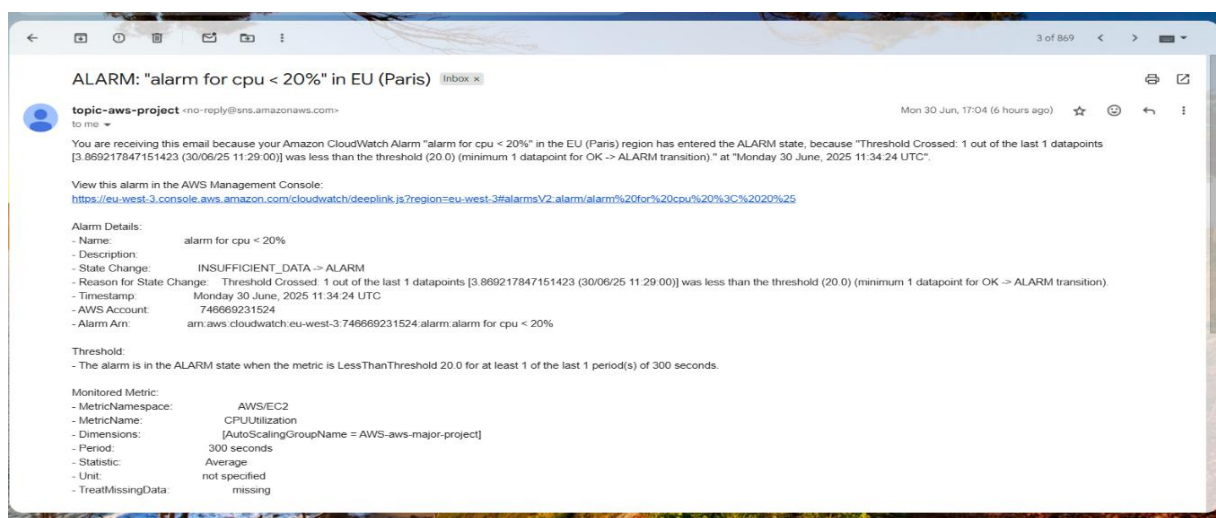
SNS

Configured Amazon SNS to send automated alerts and notifications based on CloudWatch alarms



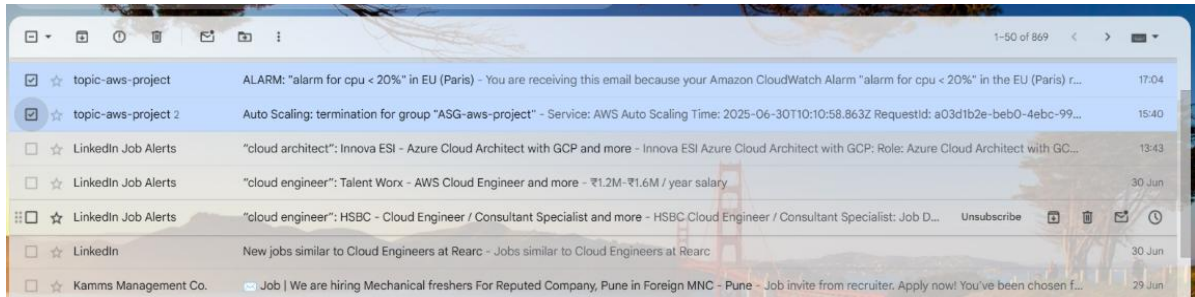
Email

email notifications for system alerts triggered by CloudWatch alarms



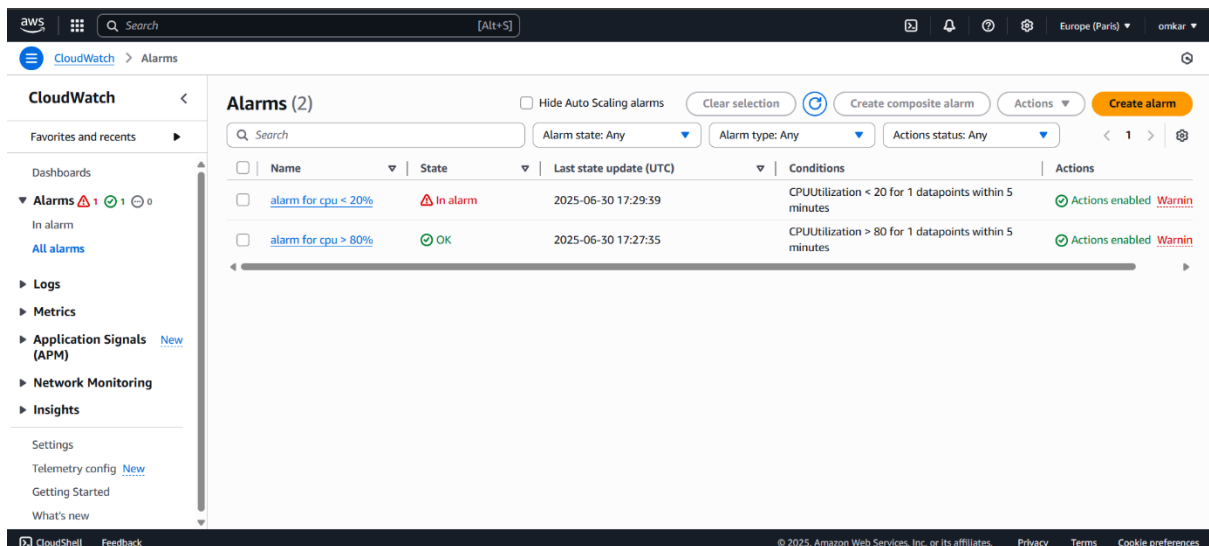
Email for Alarms

Received automated email notification via SNS triggered by a CloudWatch alarm when EC2 CPU utilization dropped below 20%.



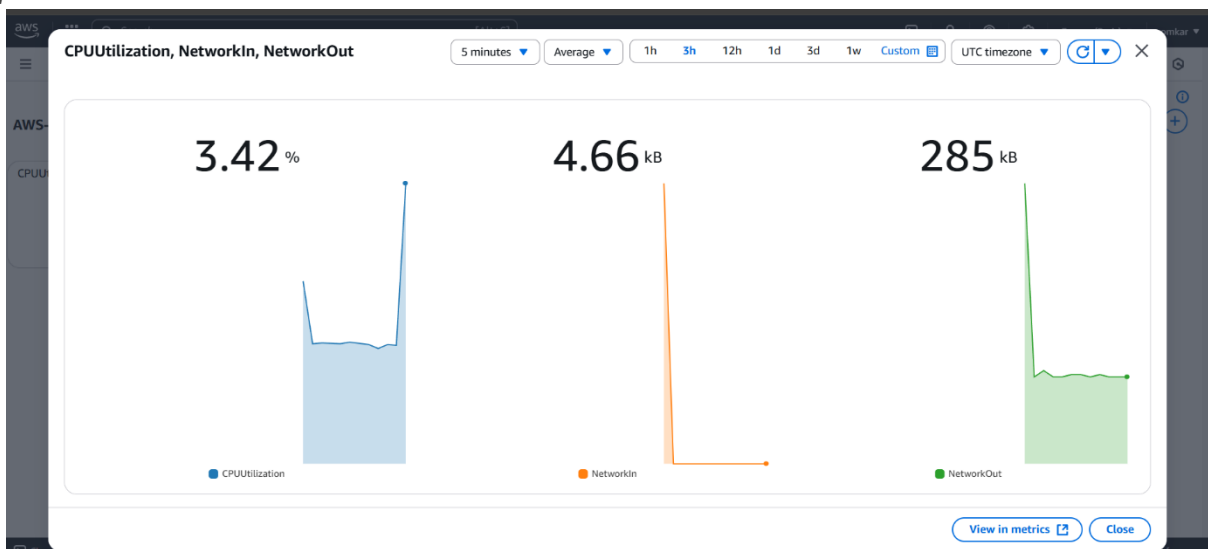
CloudWatch Alarms

CloudWatch alarm transitioned between 'In Alarm' and 'OK' states based on CPU utilization thresholds.



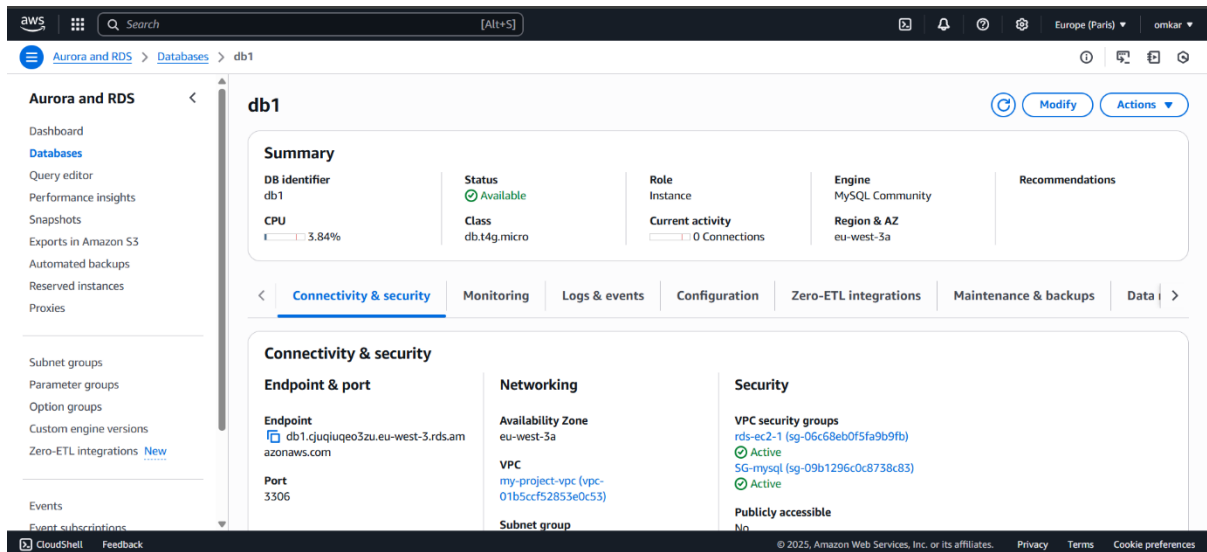
CloudWatch Dashboard

Created a CloudWatch dashboard to visualize and monitor real-time metrics of AWS resources in one place.



Aurora and RDS

Designed and deployed a secure database architecture using Amazon RDS and Aurora in private subnets to support high availability, scalability, and automated management for the project.



Databases

Created two databases, DB1 and DB2, on Amazon RDS and Aurora, and inserted structured sample data.

```
MySQL [db1]> show databases;
+-----+
| Database |
+-----+
| db1      |
| db2      |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
```

Db1

```
MySQL [db1]> SELECT * FROM contact_messages;
```

id	full_name	email	subject	message	submitted_at
1	Rahul Sharma	rahul@example.com	Web Design Inquiry	I want a website like Clean Work.	2025-06-30 13:15:15
2	Anita Mehta	anita@example.com	Partnership	Looking to collaborate on future projects.	2025-06-30 13:15:15

2 rows in set (0.00 sec)

```
MySQL [db1]> SELECT * FROM portfolio_projects;
```

project_id	project_title	category	image_url	project_description
1	Corporate Website	Web Design	images/project1.jpg	A responsive corporate website for a client.
2	Mobile App UI	App Design	images/project2.jpg	Designed UI/UX for a mobile application.
3	eCommerce Platform	Development	images/project3.jpg	Built an eCommerce backend using Laravel and MySQL.

Db2

Gg

```
MySQL [db2]> SELECT * FROM branches;
```

branch_id	branch_name	city
1	Andheri Branch	Mumbai
2	Koregaon Branch	Pune
3	Panaji Branch	Goa

3 rows in set (0.00 sec)

```
MySQL [db2]> SELECT * FROM accounts;
```

account_id	customer_id	branch_id	account_type	balance	opened_on
1	1	1	Savings	120000.50	2023-02-10
2	2	2	Current	450000.00	2023-04-01
3	3	3	Savings	78000.75	2023-06-15

3 rows in set (0.00 sec)

```
MySQL [db2]> SELECT * FROM transactions;
```

transaction_id	account_id	txn_date	txn_type	amount	description
1	1	2024-06-01	Credit	20000.00	Salary credited
2	1	2024-06-05	Debit	5000.00	Rent payment
3	2	2024-06-10	Debit	15000.00	Office expense
4	3	2024-06-12	Credit	10000.00	Freelance payment

4 rows in set (0.00 sec)

```
MySQL [db2]> SELECT * FROM loans;
```

loan_id	customer_id	loan_type	amount	interest_rate	start_date	duration_years	status
1	1	Home Loan	2500000.00	7.25	2022-05-01	20	Ongoing
2	2	Car Loan	850000.00	8.50	2023-07-10	5	Ongoing
3	3	Education Loan	500000.00	10.00	2024-01-01	7	Approved

THANK YOU